Speech Emotion Recognition Using CNN-BiLSTM for Early Depression Risk Analysis

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***Abstract*—Early detection of emotional distress is crucial as it can prevent escalation to mental health disorders like depression. Speech contains emotional information through changes of the voice e.g., pitch, loudness and tempo that indicate a mental state. Hybrid Deep Learning-based Speech Emotion Recognition System for Early Risk Analysis of Depression Speech signals are handled by preprocessing process followed by feature extraction which involves stages where Mel Spectrogram, Mel Frequency Cepstral Coefficients (MFCC), and Chroma characteristics are examples of acoustic features. representations are extracted and normalized . These features are then classified using a CNN–BiLSTM model to identify emotions including happy, sad, angry, neutral, and fear. The detected emotional patterns are further analyzed to estimate potential depression risk, providing a supportive tool for early mental health monitoring.**

***Index Terms*—Speech Emotion Recognition, CNN-BiLSTM, MFCC, Deep Learning, Depression Risk Analysis**

I. Introduction

Depression, one of the most prevalent mental health disorders in the world, severely impairs quality of life and causes impairment globally [1], [2]. which frequently goes misdiagnosed because of social stigma, ignorance, and restricted access to mental health services. [1]. Early identification of depression risk is therefore essential to enable timely intervention and prevent severe psychological consequences.

Human speech carries significant emotional information through variations in tone, pitch, speech rate, and intensity [4], [6]. These vocal characteristics often reflect an individual's emotional and psychological state. Individuals experiencing depressive symptoms frequently exhibit prolonged sadness, reduced emotional variability, and changes in vocal expression [9], [12]. Because speech is a natural and non-invasive form of communication, speech-based emotional analysis has emerged as a promising approach for early mental health assessment [2], [3].

Conventional SER methods used machine learning classifiers like Support Vector Machines (SVM) in conjunction with manually created acoustic characteristics.Random Forest Models [5], [8]. Although these techniques worked quite well, they struggled to capture the complex temporal patterns present in voice signals [4].

Despite these advancements, several challenges remain in applying speech emotion recognition for depression risk analysis [2], [3], [9], [12]. Emotional overlap between normal sadness and depressive symptoms can make classification difficult [1]. Additionally, subtle emotional variations over time are often hard to detect, particularly when working with limited emotional speech datasets.

Speech emotion recognition that mixes CNN and BiLSTM networks in order to overcome these difficulties [6], [7], [14]. The model is intended to capture the temporal and spatial properties of speech signals. Preprocessing, feature extraction, and deep learning-based emotion categorization are the methods used by the system to process speech inputs. The RAVDESS emotional speech dataset [10] is used to assess the suggested model. Although the dataset does not explicitly label depression, emotional patterns such as prolonged sadness and negative affect can provide indicators for early depression risk assessment [9], [12].

II. Related Work

Speech Emotion Recognition (SER) has recently emerged as an important research area because of its wide-ranging applications, including affective computing, healthcare analysis, and interactive human–computer systems [4]. In the initial stages of SER development, researchers primarily relied on manually designed acoustic features extracted from speech signals. Frequently used features include Mel-Frequency Cepstral Coefficients (MFCC), pitch, signal energy, and zero-crossing rate. Usually, these features were paired with traditional machine learning methods [5], [8]. Nevertheless, these conventional methods frequently have trouble capturing the temporal dynamism found in speech data. Convolutional Neural Networks

(CNNs) have gained popularity and are now widely used in contemporary SER systems because of their capacity to automatically learn complicated feature representations because to developments in deep learning techniques.

Models based on CNN can automatically learn high-level representations from the input cut-off speech signals or spectrogram, which requires no pre-processing features [6], [7], [18]. Many researches have shown that performance could be ameliorated if the audio signals are processed into spectrograms and convolutional operations to extract features from it [6], [7]. But CNN models are often designed to capture local patterns which might not effectively model long-term temporal relationships within speech data.

Sequential models like Long Short-Term Memory (LSTM) networks and Recurrent Neural Networks (RNN) were developed to get around this restriction. Because LSTM models can learn temporal dependencies across speech frames, they are especially useful for sequential data [14], [15]. Additionally, by processing sequences in both forward and backward directions, Bidirectional LSTM (BiLSTM) networks expand this capability and capture contextual information from the entire utterance [14].

More recently, hybrid architectures combining CNN and LSTM layers have been widely explored in SER research [6], [7], [14]. In such models, CNN layers are used for extracting spatial and spectral features, while LSTM layers capture temporal dynamics. These hybrid approaches generally outperform standalone CNN or LSTM models. However, many existing methods still lack the ability to focus selectively on emotionally relevant segments of speech.

Attention mechanisms have been added to SER models in order to solve this problem. The model may focus on emotionally meaningful areas of the speech signal while lessening the impact of irrelevant information thanks to attention-based strategies that allow the model to assign different priority to different portions of the speech signal. The integration of attention mechanisms with hybrid CNN–BiLSTM architectures for multi-class emotion recognition is still an open research problem, despite encouraging advancements.

Simultaneously, speech-based depression identification has become a significant field in mental health analysis. According to studies, people who suffer from depression frequently have certain vocal traits, like monotonous speech, slower speaking rates, less energy, and less pitch fluctuation. To identify depressive states using speech data, a number of deep learning models have been developed; however, many of these methods depend on clinically annotated datasets, which are scarce.

Recent work has also explored emotion recognition as an indirect indicator of depression risk. Patterns such as persistent sadness, reduced emotional variability, and lack of positive affect are commonly associated with depressive conditions. Nevertheless, only a few studies have attempted to integrate advanced SER models with structured frameworks for early depression risk assessment [13]–[16], [20].

III. Proposed Methodology

A. System Overview

The proposed system is designed to perform Speech Emotion Recognition (SER) and utilize the detected emotional patterns for early depression risk analysis [1], [2]. The overall framework consists of multiple stages, including speech acquisition, preprocessing, data augmentation, feature extraction, feature normalization, deep learning-based classification, and emotion-based interpretation [3], [4].

First, raw voice signals are collected and preprocessed to ensure consistency in both duration and quality. The processed signals are transformed into meaningful auditory features and then fed into a hybrid deep learning model that combines Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) [6], [7]..

While the BiLSTM layer records temporal relationships in speech signals, the CNN layers are in charge of extracting spatial and spectral properties from the input data. Lastly, the model divides the input speech into various emotional groups. In order to determine the likelihood of depression based on emotional patterns such chronic sorrow, decreased emotional variability, and lack of pleasant emotions, these anticipated emotions are further investigated [9].Fig. 1 shows the overall architecture of the system.

B. Dataset Description

The proposed model is evaluated using the RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song) dataset. This dataset contains high-quality audio recordings of emotional speech performed by professional actors.

The dataset consists of eight distinct emotional classes: neutral, calm, happy, sad, angry, fearful, disgusted, and surprised [1], [2], [10]. Each audio sample is labeled with its corresponding emotion, enabling supervised learning. Although the dataset does not include direct depression labels, emotional characteristics such as prolonged sadness and reduced expression of positive emotions are used as indicators for early depression risk assessment.

C. Data Augmentation

To improve the robustness of the model and reduce overfitting, data augmentation techniques are applied [8], [18].

Noise injection is performed to simulate real-world noisy environments:

$$x'(t) = x(t) + \alpha n(t) \tag{1}$$

where $n(t)$ represents Gaussian noise.

Time stretching is used to modify speech speed without affecting pitch. These techniques increase data diversity and improve generalization.

D. Feature Extraction

Feature extraction is performed to capture meaningful information from speech signals. Instead of using raw audio, several acoustic features are extracted.

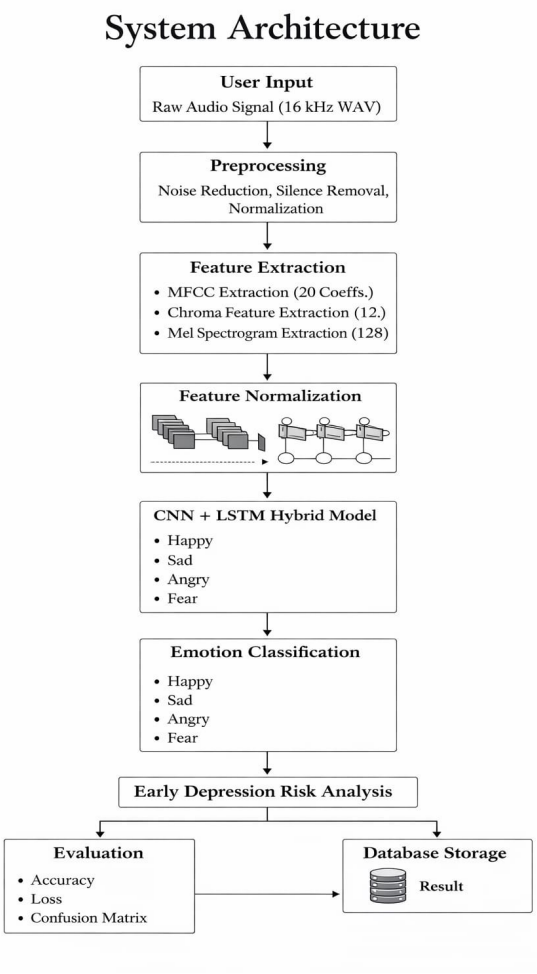


Fig. 1. Overall system architecture of the proposed framework.

MFCC are used to represent the spectral characteristics [5] of speech based on human auditory perception. In addition, delta and delta-delta coefficients are computed to capture temporal variations in speech.

Chroma features are extracted to represent pitch-related information, while Mel spectrogram features provide a time-frequency representation of the signal [6], [18].

E. Normalization of Features

Normalization is used to scale the features following feature extraction:

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma} \tag{2}$$

This enhances convergence and guarantees that every feature contributes evenly during model training.

F. Emotion-Based Depression Risk Analysis

Although the dataset does not include explicit depression labels, emotional patterns can provide useful insights into mental health.

Negative emotions are considered potential indicators of depression [1], [2], [9], [11]. A simple risk score is calculated based on the proportion of negative emotions:

$$\text{Risk} = \frac{\text{Negative Emotions}}{\text{Total Emotions}} \tag{3}$$

A higher risk score indicates a higher likelihood of depressive tendencies.

IV. Experimental Results and Discussion

A. Experimental Setup

The data is split into training and testing sets in an 80:20 ratio using stratified sampling to maintain class balance.categorical cross-entropy loss for up to 50 epochs with a batch size of 32.

B. Training Performance

The model's performance during training is examined using accuracy and loss curves.

The model effectively learns patterns from the training data, as seen in Fig. 3, where the training accuracy 95%.

The behavior of training and validation loss is also depicted in Fig. 4. the training loss continuously lowers across epochs.

C. Confusion Matrix Analysis

As presented in Fig. 5, most predictions are concentrated along the diagonal, indicating correct classification for the majority of samples. The model shows strong performance in identifying emotions such as happy, angry, and surprised.

However, some confusion is observed between similar emotional categories. For example, calm is sometimes misclassified as neutral, while fearful may overlap with sad. These errors are mainly due to similarities in acoustic properties such as pitch, tone, and speech intensity.

D. Model Comparison

A comparison is made with baseline models.

TABLE I
Performance Comparison of Models

Model	Accuracy	F1-Score
CNN	75%	0.74
LSTM	77%	0.76
CNN + LSTM	79%	0.78
Proposed CNN + BiLSTM	**81%**	**0.80**

E. Discussion

The results of the experiment show that speech emotion recognition can be successfully achieved by combining convolutional and recurrent neural networks.

Despite achieving good performance, certain challenges remain. Emotional overlap between similar classes leads to occasional misclassification. Furthermore, the dataset size is

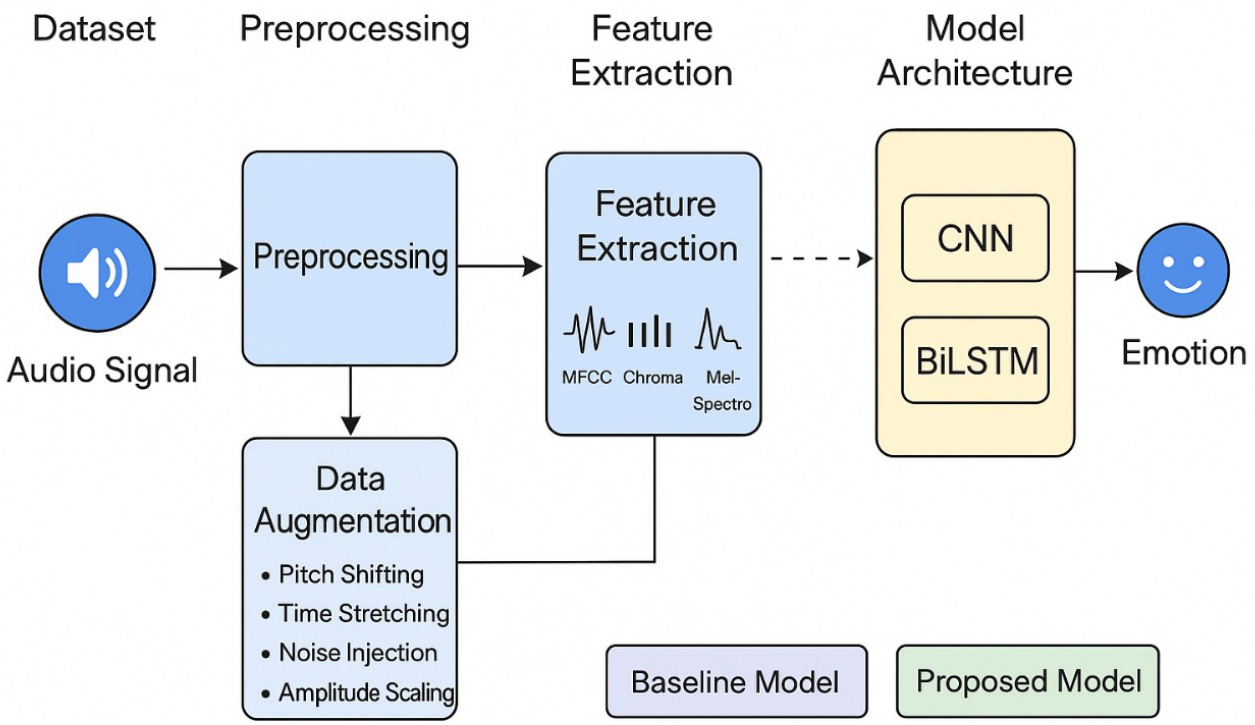


Fig. 2. Proposed CNN–BiLSTM+aug Based Speech Emotion Recognition Framework

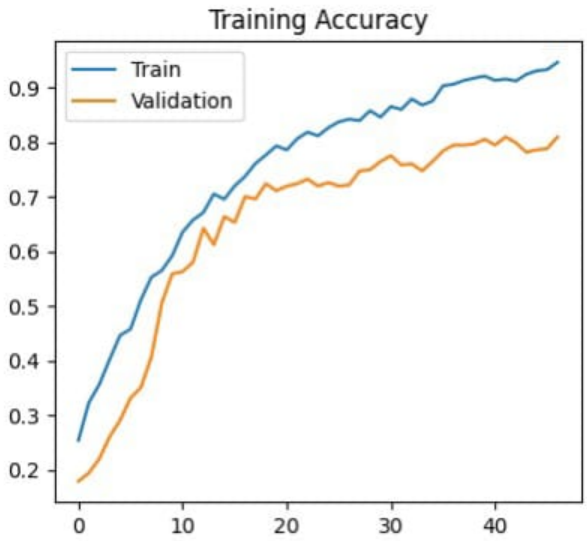


Fig. 3. Training and Validation Accuracy



Fig. 4. Training and Validation Loss

limited and may not fully represent real-world speech variability.

Subsequent enhancements may concentrate on integrating sophisticated methods like transformer-based models or attention processes.

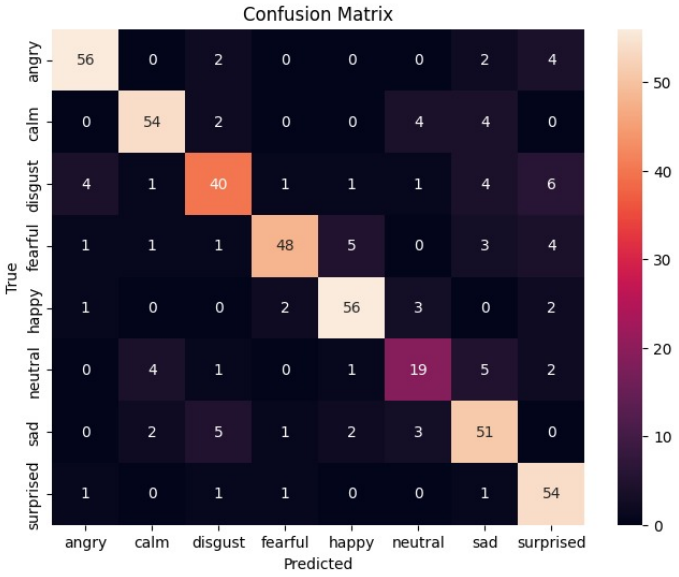


Fig. 5. Confusion Matrix of Emotion Classification

V. Conclusion

This paper presents a hybrid CNN–BiLSTM framework for speech emotion recognition and early depression risk analysis. The proposed system integrates multiple stages, including preprocessing, data augmentation, feature extraction, normalization, and deep learning-based classification, to effectively identify emotional states from speech signals. According to experimental evaluation, the model outperforms baseline deep learning models and traditional machine learning techniques, achieving an accuracy of about 81%.

Additionally, patterns of negative emotional expressions are used in an emotion-based analysis approach to estimate the likelihood of depression. This offers a straightforward but significant extension of emotion detection systems to applications including mental health monitoring.

Overall, the proposed framework demonstrates the potential of combining speech emotion recognition with depression risk assessment, contributing to the development of intelligent and accessible mental health support systems.

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